February 5th, 2026

Name and Surname .....

I, the undersigned, hereby swear that the submitted piece of work is an original piece of work and not plagiarized, copied or colluded on in way or form. In every case where the work of another has been used, it has been cited in the correct manner in the text, footnotes, index and bibliography.
With this statement, I, furthermore, confirm that I am fully aware that, in the case of plagiarism, copying or collusion there will be adverse consequences and that I am prepared to accept them.
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Definitions: Plagiarism

The unacknowledged use of another person's work either through poor academic practice or the deliberate reproduction of the work of others, including fellow students, either copied verbatim or closely paraphrased followed by the attempt to pass the work off as the students own.

Collusion

Unauthorized co-operation between at least two students with the intention of passing work off as their own individual work or where one students submits the work of another with the agreement of the latter.

☐ I accept

Business Intelligence and Data Warehouse – Final Exam
Professor E. Gallinucci
Duration: 60 minutes

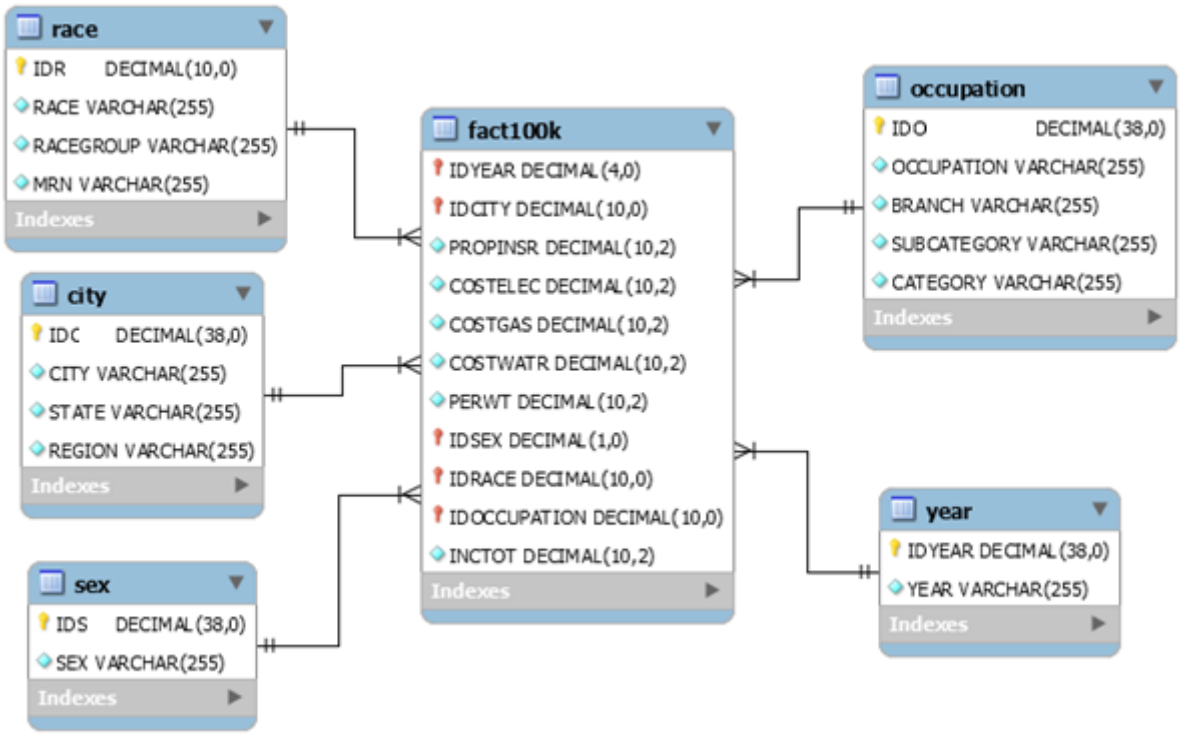
Submission rules:
Save the PowerBI workbook as a .pbix file called LASTNAME_FIRSTNAME.pbix and upload it to Moodle.

Questions:

Connection and Setup

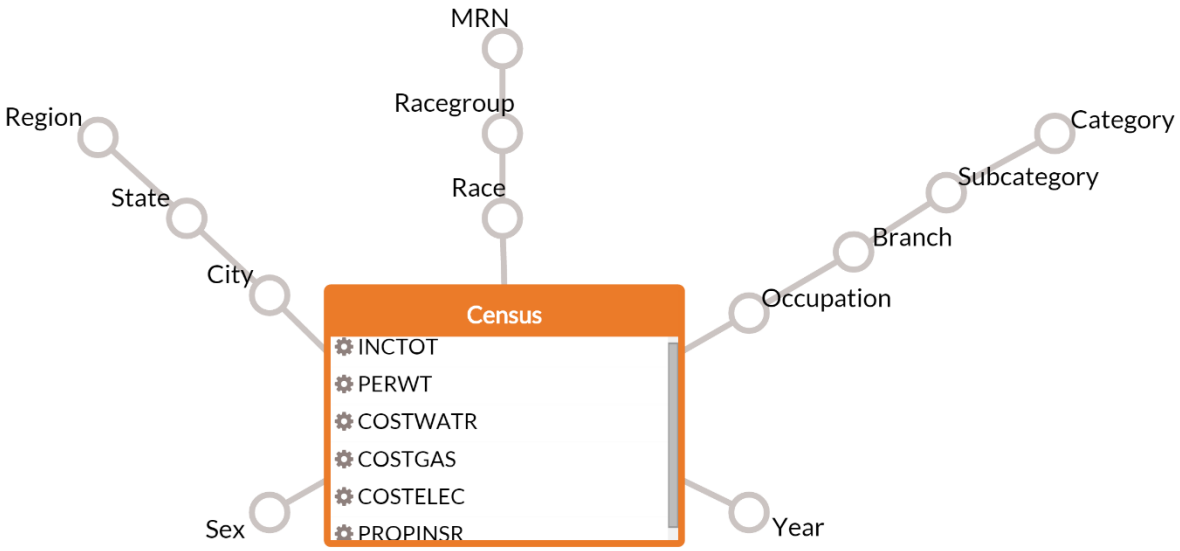
Connect to Moodle and download the CSV files in the *Exam files* folder.

Open PowerBI and connect to the CSV files provided; they contain the data of the *ipums* database, which is a sample of U.S.A. census data. Select the correct text file properties and setup the appropriate joins between the files by following the schema given in the following figure.



After the connection phase, you need to setup the resulting dimensions and measures as shown during the lectures; in particular, follow the DFM shown in the following figure to keep only the attributes shown in the DFM and to build the hierarchies.

Notes:
- PROPINSR indicates the household's annual property insurance costs
- COSTELEC, COSTGAS, and COSTWATR indicate annual costs for each housing unit
- INCTOT indicates total pre-tax personal income or losses
- PERWT indicates how many people in the U.S. population are represented by a specific row



Visualizations

Using the created PowerBI workbook, create the following visualizations.

1. We want to check the gender pay gap across occupations. Create a WEIGHTED_INCTOT field that divides the INCTOT by the PERWT. Then, create a clustered bar chart to visualize the average value of WEIGHTED_INCTOT by occupation subcategory (*Subcategory*) and *Sex*. Make sure that census data where PERWT=0 are not considered.
2. Create a pie chart to check the distribution of the sum of PERWT by *Racegroup* in the *Year* 2005.
3. Create a line chart to show the yearly (*Year*) trend of the average cost of electricity (COSTELEC), gas (COSTGAS), and water (COSTWATR); show this chart for every *Region*.
4. Create a grouping of the cost of electricity bills (COSTELEC) of size 1000. Then, create a bar chart showing, for each *Group of COSTELEC*, the average total income (WEIGHTED_INCTOT). Again, make sure that census data where PERWT=0 are not considered. Bars should be colored by average value of gas bills (COSTGAS).

General instructions

- Place each visualization in a separate page
- Intermediate steps are not required (only the final result will be evaluated)
- **For each visualization, add a textual note (using a Text box) to comment on the result** (e.g., do you observe any interesting trend/pattern?)